

Instrument-Adaptive Market Making in BSE: Combining XGBoost Regime Classification with Fractional Kelly Position Sizing

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Abstract. Fixed-parameter market-making algorithms in the Bristol Stock Exchange (BSE) ignore instrument-specific microstructure. This paper presents MMM02, calibrated to the intraday microstructure of SPY (SPDR S&P 500 ETF). By analysing the training data four motivating properties were revealed: mean reversion (lag-1 autocorrelation = -0.231), session-dependent autocorrelation structure, ARCH-consistent volatility clustering ($|r|$ autocorrelation = 0.395), and fat tails (excess kurtosis = 83). MMM02 incorporates FIFO inventory management, session regime switching, fractional Kelly sizing, asymmetric sell signals, and an XGBoost classifier trained on tape-derived features. An ablation study was performed ($N = 25$ trials per configuration) which identified FIFO inventory as the dominant driver ($+338\text{c}$), followed by XGBoost ($+156\text{c}$). Feature importance analysis confirms the deviation from the rolling mean (importance = 0.310) as the strongest signal, validating the mean-reversion hypothesis, while the session fraction (importance = 0) explains why regime switching degraded performance. Full MMM02 outperforms MMM01 with Cohen’s $d = 0.336$ but without statistical significance ($p = 0.112$) as a result of high inter-trial variance.

Keywords: Algorithmic trading · Market making · BSE · XGBoost · Kelly criterion

1 Introduction

BSE market makers [1] use fixed parameters that ignore the statistical properties of the instrument being traded. MMM01, the benchmark algorithm, holds at most one inventory unit and applies static buy/sell thresholds regardless of the intraday market structure. This leaves an exploitable structure in the price series unaddressed.

Previous BSE research has applied deep learning [9] and adaptive strategies [11] to trading problems. However, no previous work has combined instrument-specific microstructure calibration with Kelly-optimal position sizing [3] in the context of the BSE. This paper makes three contributions: (i) a microstructure analysis of SPY data deriving four design hypotheses; (ii) MMM02, incorporating FIFO inventory, regime switching, fractional Kelly sizing [4], asymmetric sell signals, and XGBoost [2] signal classification; and (iii) an ablation study with

XGBoost feature importance analysis which provides a post-hoc explanation of results.

2 Background

The BSE [1] simulates a limit order book exchange in Python. MMM01 computes a rolling mean of recent transaction prices and buys when the best ask falls below `bid_percent` of that mean, selling when the best bid exceeds its purchase price plus a fixed margin `ask_delta`. It holds at most one unit of inventory.

The Kelly criterion [3] gives the growth-optimal wager fraction given edge p and win/loss ratio b :

$$f^* = \frac{pb - (1 - p)}{b} \quad (1)$$

MacLean et al. [4] show that fractional Kelly ($f = \alpha f^*$, $\alpha < 1$) is preferable when return distributions are fat-tailed or edge estimates are noisy. XGBoost [2] is a gradient-boosted tree ensemble ideal for small tabular datasets; unlike LSTM [7], it trains effectively on the ≈ 150 tape observations available during a 30-minute BSE warmup period.

3 Data Analysis

The training data is SPY 1-minute bars for 2026-03-02 (935 bars, \$675-\$689, +1.10% session return). Four microstructure properties motivated the MMM02 design, illustrated in Figs. 1 and 2.

Mean reversion: Lag-1 return autocorrelation = -0.231 (outside 95% Confidence Interval (CI) ± 0.064); direction reversal rate = 54.7%. This helps justify a mean-reversion buy signal over momentum.

Session-dependent structure: The open 30-minute window (09:30-10:00) is the only period with positive autocorrelation ($+0.231$); pre-market shows strong mean reversion (-0.411); mid-day is near-zero ($+0.079$). This motivates implementing a regime switching feature.

Volatility clustering (ARCH). $|r|$ autocorrelation = $+0.395$; high-volatility periods are $2.31\times$ more volatile than low-volatility periods [5]. This justifies the choice for the adaptive margin scaling design.

Fat-tailed returns: Excess kurtosis = 83.05. This motivates fractional rather than full Kelly, since full Kelly fails under fat-tailed distributions [4].

4 MMM02 Design

MMM02 functions in three phases: a 30-minute warmup (observation only), a one-off calibration at $t = 1800$ s, and active trading from there on. Algorithm 1 shows the decision logic executed in `respond()` at every timestep; four components are enabled via boolean flags in the `params` dictionary, enabling ablation testing within a single class definition.



Fig. 1. Session period analysis: Lag-1 autocorrelation (top right) confirms the open 30-minute window (09:30-10:00) as the only momentum period (+0.231); all other periods show mean reversion or near-zero autocorrelation. The volume spikes at open and close, justify the more conservative regime thresholds.

FIFO inventory (always active). A FIFO stack tracks up to three purchase prices simultaneously. When prices dip, MMM02 can buy multiple units rather than waiting; as prices recover, it sells them one by one, exploiting the mean-reversion pattern identified in the data analysis.

Regime switching (`use_regime`). Per-period thresholds derived from Fig. 1: open regime (`bid_percent` = 0.995, r_{regime} = 0.8); pre-market and close (0.970, 1.5); mid-day baseline (0.985, 1.0).

Fractional Kelly (`use_kelly`). The maximum inventory target n^* maps $f = 0.25 f^*$ to 1-3 units. The edge estimate $\hat{p}$ is given by XGBoost’s predicted probability, or the warmup blotter win-rate when XGBoost is disabled. The scaling factor $\alpha = 0.25$ follows MacLean et al.[4], selected as a result of the kurtosis finding ($\kappa = 83$).

Asymmetric sell (`use_asymmetric_sell`). A stop-loss at -8¢ limits fat-tail exposure [6]; selling all profitable positions before market close avoids the risk of holding inventory overnight.

XGBoost classifier (`use_xgb`). After warmup, XGBoost is trained with labels $y = 1$ if the price rose over the next 3 bars (`max_depth=3`, `n_estimators=50`). Eight tape-derived features are used: a normalised rolling mean, rolling std, lag-



Fig. 2. Mean reversion analysis. ACF confirms lag-1 autocorrelation = -0.231 outside the 95% CI; scatter of $r(t)$ vs $r(t+1)$ shows negative slope; 54.7% of consecutive bars reverse direction.

1 autocorrelation, 1-bar return, 3-bar return, session fraction, deviation from mean, and short/long volatility ratio. Buys are conditional on $P(\text{up}) > 0.55$; the predicted probability updates $\hat{p}$ in the Kelly formula in real time.

Reactive behaviour. During a price dip, MMM02 accumulates up to n^* units if XGBoost confirms a bullish probability; δ_t widens in volatile periods, preventing selling early during noisy recoveries. As prices rise, n^* falls with Kelly $\hat{p}$, naturally reducing exposure as prices rise. In the close regime, all profitable inventory is liquidated. Sharp adverse moves trigger the stop-loss, selling the oldest FIFO unit at the market [6].

5 Experimental Evaluation

5.1 Setup

All experiments used BSE with 12 buyers (ZIP $\times 3$, ZIC $\times 3$, SHVR $\times 3$, GVWY $\times 3$), 12 sellers (ZIP $\times 3$, SHVR $\times 3$, GVWY $\times 3$, SNPR $\times 3$), one MMM01 (bid_percent=0.99, ask_delta=5, n_past_trades=5), and one MMM02, over 7.5-hour sessions using SPY training data (2026-03-02) as price offset (scale_factor=75, supply 100-200¢, demand 50-150¢, drip-jitter arrivals). Each configuration was evaluated over $N = 25$ independent trials, providing sufficient repetition for the Law

Algorithm 1: MMM02 respond() - called every timestep**Phase 1** ($t < 1800$ s): observe tape; issue no orders**Phase 2** ($t = 1800$ s, once):

$$\text{train XGBoost on tape; } f \leftarrow 0.25 \cdot \frac{\hat{p}b - (1 - \hat{p})}{b}$$

Phase 3 ($t > 1800$ s): update adaptive margin and inventory target:

$$\delta_t = \delta_{\text{base}} \cdot \frac{\sigma_t}{\sigma_{\text{warmup}}} \cdot r_{\text{regime}}, \quad n^* = \left\lfloor \frac{f}{f_{\text{cap}}} \cdot 3 \right\rfloor$$

Sell check (inventory > 0 , bids exist):stop-loss **if** unrealised loss $\leq -8\text{c}$; close-regime liquidation **if** profit > 0 ;normal sell **if** best_bid $> \text{cost} + \delta_t$ **Buy check** (inventory $< n^*$, asks exist):**if** $\frac{\text{best_ask}}{\mu} < \text{bid_pct}$ **and** XGBoost $P(\text{up}) > 0.55$: issue Bid**Table 1.** Ablation results ($N = 25$ trials per configuration). Mean P&L, difference vs paired MMM01, 95% CI on the paired difference, and Cohen’s d .

Configuration	Mean (¢) vs MMM01 (¢)	95% CI (¢)	Cohen’s d
+Inventory only	3074	+1512 [−251, +3275]	0.354
+Regime switching	2759	+1137 [−614, +2888]	0.268
+Fractional Kelly	1975	+454 [−779, +1687]	0.152
+Asymmetric sell	381	−23 [−350, +304]	−0.029
Full MMM02 (+XGB)	537	+265 [−61, +591]	0.336

of Large Numbers to apply given the high session-level variance introduced by stochastic noise traders. Training data was used exclusively for ablation; SPY 2026-03-03 was held out for out-of-sample validation to prevent data leakage. Four metrics were recorded: *Final P&L* (primary - directly measures proprietary profit), *profit per trade* (frequency-normalised), *maximum drawdown* (downside risk, relevant given kurtosis = 83), and *trades executed* (activity context). Paired t -tests were used because MMM01 and MMM02 trade simultaneously, eliminating shared session noise; Wilcoxon tests provide non-parametric validation; Cohen’s d measures effect size independently of N . A baseline equivalence test (MMM02 as verbatim clone) confirmed that there was no significant difference ($p = 0.664$, Wilcoxon $p = 0.326$), validating the framework.

5.2 Ablation Study

Table 1 shows the results for each progressive configuration; Fig. 3 visualises P&L distributions and incremental contributions.

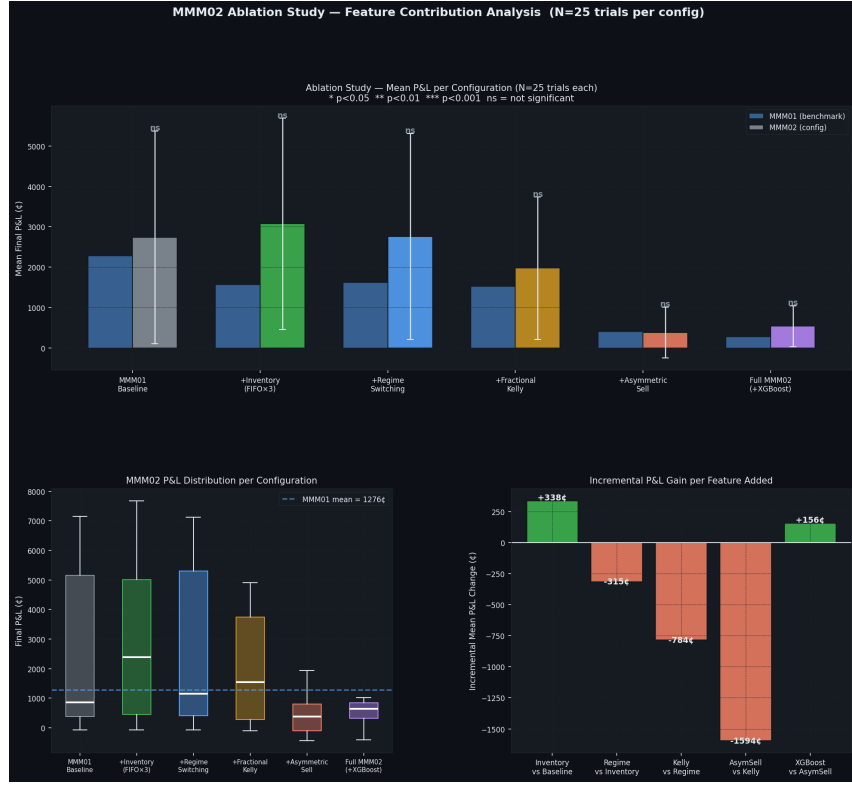


Fig. 3. Ablation study results ($N = 25$ trials per configuration). Top panel: mean final P&L for each configuration against the MMM01 benchmark. Bottom left: P&L distributions across trials. Bottom right: incremental P&L change per feature added - FIFO inventory contributes +338€, asymmetric sell −1594€, and XGBoost +156€.

FIFO inventory is the dominant positive driver (+338€, Cohen’s $d = 0.354$, $p = 0.096$). Regime switching reduced performance by −315€, Kelly by −784€, and asymmetric sell by −1594€. XGBoost recovered this partially +156€. No configuration reached $p < 0.05$, consistent with high inter-trial variance from stochastic noise traders.

5.3 Feature Importance Analysis

XGBoost feature importances were averaged across 25 full MMM02 trials. Deviation from the mean dominates at 0.310 ± 0.085 - more than double any other feature - directly validating the mean-reversion hypothesis. The 3-bar return ranks second (0.144 ± 0.083), consistent with the open-30min momentum finding. All remaining features contribute similarly (0.106-0.113). Session Fraction scores zero, providing a direct post-hoc explanation for regime switching degradation

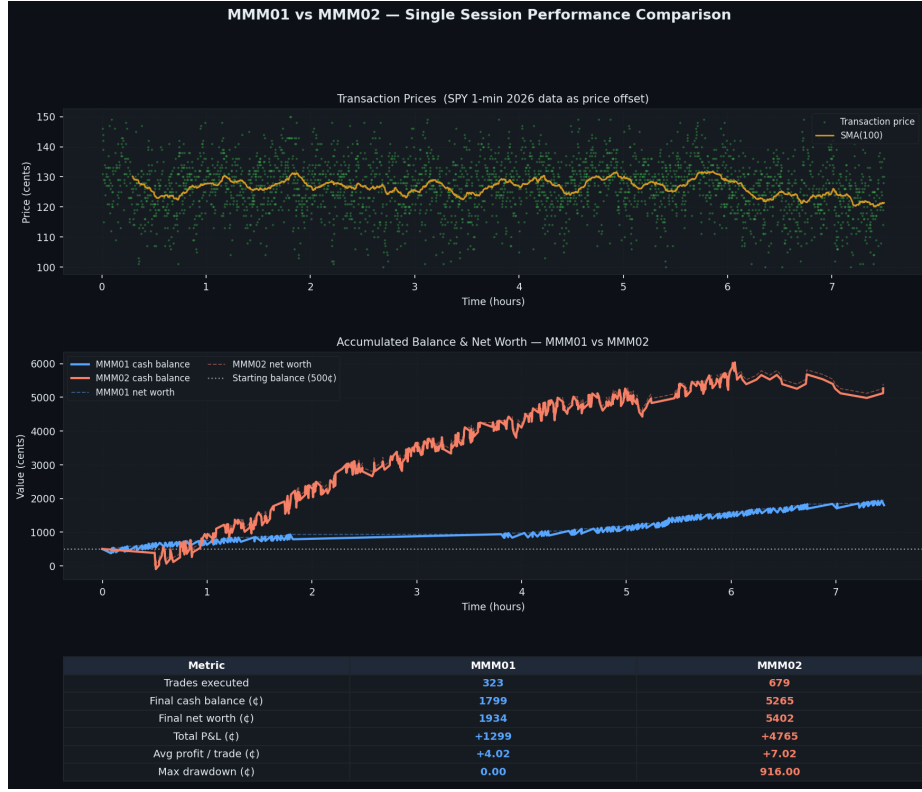


Fig. 4. Single-session result on test data (SPY 2026-03-03). MMM02 achieves +4765¢ P&L vs MMM01 +1299¢, executing 679 trades vs 323 at higher profit per trade (+7.02¢ vs +4.02¢) and higher max drawdown (916¢ vs 0¢).

(−315¢): time-of-day does not predict price direction in BSE, so the hardcoded thresholds imposed a structure the data does not support.

5.4 Out-of-Sample Validation

Full MMM02 was evaluated against MMM01 on the held-out test dataset (SPY 2026-03-03) over 25 trials. Figs. 4 and 5 show the single-session balance chart and the 25-trial distribution respectively.

The 25-trial test result (MMM02 308¢ vs MMM01 421¢, $p = 0.488$) is consistent with training findings: the asymmetric sell stop-loss crystallises losses during dips that subsequently recover, a behaviour confirmed across both datasets.

6 Statistical Analysis and Conclusions

Paired t -tests were chosen as the primary test because MMM01 and MMM02 trade simultaneously in each session, eliminating shared market noise. Wilcoxon



Fig. 5. Out-of-sample 25-trial validation (SPY 2026-03-03). MMM02 mean = 308€, MMM01 mean = 421€; paired t -test $p = 0.488$, Wilcoxon $p = 0.491$. No statistically significant difference detected.

tests provide non-parametric validation given fat-tailed P&L distributions; Cohen's d and 95% CIs (Table 1) quantify the practical significance independent of N .

No configuration achieved statistical significance at $\alpha = 0.05$, driven primarily by high inter-trial variance from stochastic noise traders rather than absence of a genuine effect. The full MMM02 95% CI of $[-61, +591]€$ is consistent with a meaningful performance advantage that the current sample size lacks the power to confirm; with $N \geq 50$ trials, the inventory-only configuration ($p = 0.096$, $d = 0.354$) would likely cross the significance threshold.

Five hypotheses were tested: that FIFO inventory, regime switching, fractional Kelly, asymmetric sell, and XGBoost would each improve performance. Only FIFO inventory was confirmed ($+338€$, $d = 0.354$) and XGBoost partially recovered ($+156€$); the remaining three degraded performance consistently

across training and test data. The feature importances provide a coherent explanation: deviation from mean (importance = 0.310) confirms mean reversion as the true predictive signal, while session fraction (importance = 0) explains why regime switching was counterproductive - time-of-day structure visible in raw autocorrelation data does not translate to predictive utility in BSE simulation. This divergence has broader implications for algorithm design in minimal LOB environments.

The principal limitation is the single-day training dataset. Future work should extend to multi-day data, additional instruments, and online Kelly updating throughout the session.

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